ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): the design process and specifications, a two-cylinder pneumatic box lifter, displacement-step diagrams, pneumatic force calculations, reed switches and flow control, electro-pneumatic ladder diagrams, bills of materials, Gantt charts, saving compressed air, commissioning, fault-finding and maintenance.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Take g = 9,81 m/s².
5. Draw sketches, charts and circuits neatly. Use only a black or blue pen.

> **PROJECT.** In a warehouse, boxes arrive on a lower conveyor and must be moved to an upper conveyor 600 mm higher. Workers lift them by hand, which causes back injuries. The manager wants a pneumatic box lifter: cylinder A lifts a platform with the box (A+), cylinder B pushes the box onto the upper conveyor (B+), then the platform goes down (A−) and cylinder B returns (B−). Reed switches a0, a1, b0 and b1 show the end positions of the cylinders, and each cylinder is controlled by a 5/2-way double-solenoid valve.

QUESTION 1

1.1 Name and explain FOUR stages of the design process. (4)

1.2 Write SIX specifications for the box lifter. (6)

[10]

QUESTION 2

2.1 Make a neat freehand sketch of the box lifter. Show the lower and upper conveyors, cylinder A with the platform, cylinder B and the reed switches. (5)

2.2 Draw the displacement-step diagram for the sequence A+ B+ A− B−. (6)

2.3 Cylinder A has a bore of 63 mm and works at an air pressure of 600 kPa. Friction losses are 10%. Calculate the:

2.3.1 theoretical force of cylinder A when it extends (2)

2.3.2 effective (useful) force of cylinder A (2)

2.3.3 largest mass of a box that can be lifted if the platform has a mass of 15 kg (2)

2.4 Explain the function of a one-way flow control valve fitted to cylinder A. (2)

2.5 Explain how a reed switch detects the position of the piston. (3)

2.6 Explain why the exhaust air (meter-out), rather than the supply air, is usually throttled to control cylinder speed. (3)

[25]

QUESTION 3

3.1 Draw a ladder diagram that controls the sequence A+ B+ A− B− with a START button and the reed switches a0, a1, b0 and b1. Solenoids: Y1 (A+), Y2 (A−), Y3 (B+) and Y4 (B−). (8)

3.2 Draw up a bill of materials for the box lifter with suitable column headings and at least SIX items. (8)

3.3 Draw a Gantt chart for SIX activities of the project over a period of 5 weeks. Some activities must overlap. (10)

3.4 Give FOUR ways to reduce the cost of the compressed air used by the box lifter. (4)

[30]

QUESTION 4

4.1 Use a spider diagram to show SIX areas that must be tested on the box lifter. (6)

4.2 Explain how the speed of the cylinders is set during commissioning. (3)

4.3 Explain why the air must be switched on slowly (with a soft-start valve) the first time. (2)

4.4 List FOUR items that must be included in the commissioning report. (4)

[15]

QUESTION 5

5.1 Draw up a maintenance schedule for the box lifter with FOUR tasks and how often each must be done. (8)

5.2 Cylinder B does not extend, although the PLC output LED for Y3 is on. Describe a logical fault-finding procedure. (6)

5.3 Explain how an air leak in the system can be found. (2)

5.4 Explain why water must be drained from the air filter regularly. (2)

5.5 Give TWO reasons why spare seal kits for the cylinders should be kept in stock. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER